The Relay — Final Engineering Report

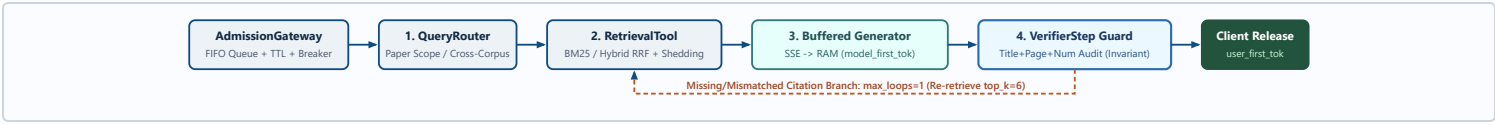
Grounded Multi-Step Research Pipeline & Load-Resilient Gateway • Vaani Research Assignment

Repo: `github.com/abo7adeed/The-Relay` • Python 3.13 • `109/109` Tests Passing

1. EXECUTIVE SUMMARY

The Relay is a four-step Python/FastAPI research assistant answering engineering questions strictly from three exact arXiv papers: *Attention Is All You Need* (`1706.03762v7`), *Moshi* (`2410.00037v1`), and *H-Net / Dynamic Chunking* (`2507.07955v2`). Every answer must cite `[Paper Title, Section Title, Page X]` or refuse safely when evidence is absent. Under bursty production workloads hitting strict free-tier TPM/RPM limits, naive sequential pipelines leak unverified hallucinations while streaming and collapse under retry storms. The Relay places a bounded FIFO `AdmissionGateway` in front of the pipeline, buffers SSE drafts in memory until `VerifierStep` approves them, sheds retrieval depth near quota exhaustion, and trips a zero-retry `CircuitBreaker` when daily budget is exhausted.

2. ARCHITECTURE & REQUEST LIFECYCLE



- **Admission & Deadline Guard (`gateway.py`):** Bounded `asyncio.Queue(maxsize)` drained by `C` workers. Overflow rejects immediately (`429` `rate_limited`, `0` tokens). Dequeued calls record `queue_wait_ms`, fast-fail (< 3 ms) if `CircuitBreaker.is_open()`, and drop expired `ctx.deadline_mono` calls (`timeout`).
- **Routing & Tool Execution (`router.py`, `tool.py`):** Single-paper entity matches scope search to one paper; cross-paper/general queries search all three papers. `RetrievalTool` cuts `effective_k -> max(2, effective_k // 2)` when local quota pressure ≥ 85%.
- **Buffered Generation & Verification Loop (`generator_step.py`, `verifier_step.py`):** SSE deltas buffer in RAM (`temperature=0.0`, `max_tokens=600`), measuring `model_first_token_ms`. `VerifierStep` audits paper title prefix + exact integer page equality (blocking page-substring false positives like Page `1` vs `10`) and verifies numeric tokens (≥ 10 or decimals). On missing/mismatched citations with TTL slack, `PipelineEngine` retries retrieval once (`top_k=6`); unverified output is suppressed. `PipelineConfig.enforce_verification_always=True` is validation-locked.

3. RETRIEVAL DESIGN & CONFIGURATION COMPARISON

Offline PyMuPDF parsing (`parser.py`, `chunker.py`) removes table-header noise (e.g., `Model`, `Background`), repairs line-break hyphens conservatively without breaking technical compounds, and preserves physical PDF page numbers across three settings:

Setting ID	Chunk Spec / Corpus Size	Retrieval & Rerank Architecture	Full Facts %	Cite OK %	Tok / Ans
`setting_1_bm25_base`	350w / 50o (297 chunks, 217.9 avg w)	`BM25Okapi`, `top_k=4`, no reranker	53.3%	63.6%	1900.3
`setting_2_bm25_granular`	175w / 25o (481 chunks, 139.8 avg w)	`BM25Okapi`, `top_k=6`, no reranker	46.7%	63.6%	1789.4
`setting_3_hybrid_rerank` (Chosen)	350w / 50o (297 chunks, 217.9 avg w)	`BM25Okapi` (`k=8`) + L2 TF-IDF subword cosine RRF (`top_k=3`)	**53.3%**	54.5%	**1373.1**

`HybridReranker` (`reranker.py`) fuses BM25 candidates with sparse TF-IDF cosine similarity over content tokens, 4-char subword n-grams (`#xxxx`), and section/paper headers via Reciprocal Rank Fusion without GPU overhead. Setting 3 matches Setting 1 on full-fact accuracy (`53.3%`, `33.3%` correct+cited) while cutting prompt volume by **27.7%**, whereas Setting 2 drops to `46.7%` by splitting cross-sentence facts (`Q05`: *Mimi* vs *1.1 kbps*).

4. RELIABILITY, CONCURRENCY & CACHING BEHAVIOR

- **Sliding Rate Limiter & Context Shedding (`rate_limiter.py`):** Tracks 60s rolling RPM/TPM usage; at ≥ 85% utilization, `quota_pressure=True` reduces retrieval `top_k` (`3 -> 2`) while keeping `VerifierStep` active.
- **Bounded Retries & Timer Isolation (`retry_policy.py`):** Caps calls at `max_attempts=2` (one retry), limits retries to a `15%` global budget after ≥ 5 calls, uses full jittered exponential backoff (`base=0.6s, cap=3.0s`) gated by remaining TTL and `retry-after ≤ remaining_s`, skips non-retryable `400/401/403/404` and `daily_quota_exhausted`, and isolates sleeps inside `retry_wait_ms` (`total_ms` conservation error ≤ `0.195 ms` vs `15 ms` limit).
- **Circuit Breaker (`circuit_breaker.py`):** Opens immediately on daily exhaustion (< 2 ms fast-fail, `0` tokens for `3600s` cooldown or reset) or `10` consecutive `5xx` errors.
- **Caching (Implemented vs. Absent):** Offline PDFs cache in `data/raw_papers/`, chunk corpora prebuild to JSON, BM25/TF-IDF matrices stay cached in RAM after startup, and table bboxes use `_TABLE_BBOX_CACHE` offline. **No semantic/query or LLM response cache exists on the online request path.**

5. KEY ENGINEERING DECISIONS

1. Buffered Verification Invariant vs. TTFT: Withholding SSE tokens until `VerifierStep` finishes adds ~350–610 ms post-first-token delay, trading streaming speed to guarantee zero unverified numeric claims reach callers.

2. Hybrid RRF & Strict FIFO vs. Neural Reranking / Priority Queues: Sparse subword RRF avoids GPU serving overhead; bounded FIFO prevents starvation and keeps burst tail latency deterministic without external brokers.

Experiments, Results, Live Validation & Limitations

120-Row Stand-In Matrix (`results.csv`) Kept Strictly Separate from Live Direct Groq Probes

`results.csv` SHA-256: `441b0004...c3cdd` • Strict Revision 4 Grading

1. BENCHMARK METHODOLOGY

All `120` matrix rows executed on Windows 11 (Python 3.13) over localhost HTTP/SSE against FastAPI `StandInService` (`standin_service.py`), configured with Log-Normal TTFT (μ=5.45, σ=0.35, ~233 ms base median), `6 ms`/token SSE decode, saturation curve `1 + 1.2·max(0, active/5 - 1)`, and simulated `llama-3.1-8b-instant` quotas (`30` RPM, `6,000` TPM, `500,000` TPD; cost ref `$0.05/1M` in, `$0.08/1M` out). **Retrieval Eval (45 rows):** `Q01–Q15` across Settings 1–3 under `500k` TPM override, graded via strict all-facts conjunction and boundary regex (`grading.py`). **Load Suite (75 rows, Setting 3):** `steady` (N=4, 16s gap, C=3, q=50, TTL=10s), `burst` (N=20, 0.02s gap, C=2, q=18, TTL=2.2s), `overload` (N=24, 0.15s gap, C=1, q=10, TTL=1.2s), `limit_halved` (N=12, 2s gap, C=2, q=20, TTL=4s, 15 RPM/3k TPM), and `budget_exhausted` (N=15, 0.05s gap, C=3, force daily 429).

2. QUANTITATIVE RESULTS (`RESULTS.CSV`, ALL ROWS `SOURCE=STAND-IN`)

Load Scenario	N	Dur(s)	QPM	FT = TOT p50 / p95 / p99 (ms)	OK%	429%	TO%	Retry/Req	PeakQ	Tok/Ans	$/1k Ans
`steady`	4	48.9	5.0	622.1 / 886.6 / 916.0	100.0	0.0	0.0	0.00	1	978.0	$0.0500
`burst`	20	1.9	2001.5	1064.7 / 1334.4 / 1673.4	25.0	75.0	0.0	0.00	18	1118.2	$0.0570
`overload`	24	3.6	399.7	1231.1 / 1772.0 / 1882.0	20.8	37.5	41.7	0.00	10	1132.2	$0.0579
`limit_halved`	12	22.1	32.6	11.4 / 3025.6 / 3027.3	16.7	83.3	0.0	0.17	1	1209.0	$0.0616
`budget_exhausted`	15	0.9	1041.0	0.1 / 1.7 / 4.6	0.0	100.0	0.0	0.00	1	0.0	$0.0000

Retrieval Config (`N=15` each)	Full Facts%	Cite OK%	Facts+Cite%	Wrong%	Refused OK%	Wrong Ref%	Tok/Ans	FT/TOT p50 / p95 (ms)
1: `350w / 50o, k=4, rerank=No`	53.3%	63.6%	33.3%	13.3%	26.7%	6.7%	1900.3	579.1 / 737.8
2: `175w / 25o, k=6, rerank=No`	46.7%	63.6%	33.3%	13.3%	26.7%	13.3%	1789.4	544.7 / 752.9
3: `350w / 50o, k=8->3, rerank=Yes`	**53.3%**	54.5%	33.3%	13.3%	26.7%	**6.7%**	**1373.1**	560.9 / 871.2

- **Latency Breakdown & Bottlenecks:** Buffering aligns client `first_token_ms == total_ms`. At `1,373.1` tokens/answer, `6,000` TPM saturates at `~4.4` QPM—far below `30` RPM. Slow `overload` trace `7afa5482...` took `1914.041 ms`: `1147.689 ms` queued (`60.0%`) + `766.345 ms` working (`40.0%`, ε=0.007 ms); `model_first_token_ms=1303.810 ms` proves `68.1%` preceded first token and `31.9%` (`610.2 ms`) buffered/verified afterward. Under `limit_halved`, trace `3803d90f...` slept `3008.6 ms` on `retry-after: 3s` vs `18.956 ms` work (~160x inflation) and still failed 429.

3. LIVE API VALIDATION (`DIRECT_GROQ` VS. STAND-IN)

Live probes hit Direct Groq (`https://api.groq.com/openai/v1`) via `probe_groq.py` and `probe_live_pipeline.py` with local `GROQ_API_KEY` (**strictly excluded from `results.csv`**):

Metric / Capability	Doc Ref (`llama-3.1-8b-instant`)	Live Measured (`direct_groq`, `qwen/qwen3.8-27b`)	Status & Evidence Source
Requests / min (`rpm`)	30 (docs)	**NOT VERIFIED** (`null`)	`x-ratelimit-limit-requests=1000` (`reset=1m26.4s`) maps to RPD; RPM unemitted
Tokens / min (`tpm`)	6,000 (docs)	**8,000** (`reset=269ms` ping / `6.2–13.3s` RAG)	**PASS** — HTTP header `x-ratelimit-limit-tokens` (`observed_limits.json`)
Requests / day (`rpd`)	14,400 (docs)	**1,000** (`86,400s / 1,000 = 86.4s` step)	**PASS** — HTTP header `x-ratelimit-limit-requests` (`observed_limits.json`)
Tokens / day (`tpd`)	500,000 (docs)	**NOT VERIFIED** (`null`)	**NOT VERIFIED** — Not returned in HTTP response headers
Live SSE / Scope Gaps	Stand-in: `llama-3.1-8b-instant`	`/models`: 11 models (`llama-3.1-8b-instant` absent); 1 ping `368.9 ms`; 5 SSE calls (`11s` gap): `5/5` OK, TTFT `446.2–1,392.8 ms`, `6,370` in / `465` out	**PASS** (`real_groq_probe_results.json`: `4/4` target-cited, `3` correct, `1` refused_correctly `Q14`, `1` false negative `Q02`). Live 429 burst & `RESPAN_API_KEY` gateway: **NOT VERIFIED**

4. OBSERVED LIMITATIONS & FAILURE ANALYSIS

Known Live Failure — Q02 Grading False Negative: On live request `95aa0152...` (*warmup steps*), `qwen/qwen3.8-27b` retrieved chunk `1706.03762v7_c0022` (Page 7), passed grounding (`verification_status="verified_grounded"`, `citation_grade="citation_correct"` for `[Attention Is All You Need, 5.3 Optimizer, Page 7]`), and answered **4,000** warmup steps. Unmodified Revision 4 `grade_eval_answer` classified it `refused_wrongly` because its multi-paper opening ("*The provided papers do not state a single universal number...*") matched `_REFUSAL_INDICATORS` (`"do not state"`), and `_fact_present` does not normalize thousands commas (`"4,000"` vs `"4000"`).

- **Measured vs. Unknown:** Lexical BM25/TF-IDF misses synonyms (`Q10` fails to match "*routing module*" to "*without a tokenizer*"), and Stand-in multi-fact synthesis yields `13.3%` wrong (`Q06`, `Q09`). Stand-in models LPU compute via log-normal sleeps without noisy-neighbor interference; live RPM, TPD, concurrency caps, and 429 burst recovery remain unmeasured.

5. CONCLUSION

All `109/109` offline unit/regression tests, `120`-row CSV validations (`ε < 0.2 ms`), GitHub Actions CI (Run `43099888608` on `a871a51`), and paced live Direct Groq SSE calls are verified. Production readiness requires adding an online response/query cache, dense semantic retrieval for vocabulary gaps (`Q10`), external Redis queueing, live target-model RPM/TPD stress testing, and decoupling evaluation refusal heuristics from cited answers (`Q02`).